

Solutions of Problems on ch 24

1. A flat surface of area 3.20 m^2 is rotated in a uniform electric field of magnitude $E = 6.20 \times 10^5 \text{ N/C}$. Determine the electric flux through this area (a) when the electric field is perpendicular to the surface and (b) when the electric field is parallel to the surface.

(a) The electric field is perpendicular to the surface, so $\theta = 0^\circ$:

$$\Phi_E = (6.20 \times 10^5 \text{ N/C})(3.20 \text{ m}^2)\cos 0^\circ$$

$$\Phi_E = \boxed{1.98 \times 10^6 \text{ N} \cdot \text{m}^2/\text{C}}$$

(b) The electric field is parallel to the surface: $\theta = 90^\circ$, so $\cos \theta = 0$, and the flux is zero.

3. A 40.0-cm-diameter circular loop is rotated in a uniform electric field until the position of maximum electric flux is found. The flux in this position is measured to be $5.20 \times 10^5 \text{ N} \cdot \text{m}^2/\text{C}$. What is the magnitude of the electric field?

$$E = \frac{\Phi_{\text{max}}}{A} = \frac{\Phi_{\text{max}}}{\pi r^2}$$

$$E = \frac{5.20 \times 10^5 \text{ N} \cdot \text{m}^2/\text{C}}{\pi(0.200 \text{ m})^2} = 4.14 \times 10^6 \text{ N/C} = \boxed{4.14 \text{ MN/C}}$$

7. An uncharged, nonconducting, hollow sphere of radius 10.0 cm surrounds a $10.0\text{-}\mu\text{C}$ charge located at the origin of a Cartesian coordinate system. A drill with a radius of 1.00 mm is aligned along the z axis, and a hole is drilled in the sphere. Calculate the electric flux through the hole.

$$\begin{aligned}
 \Phi_{E, \text{hole}} &= \vec{\mathbf{E}} \cdot \vec{\mathbf{A}}_{\text{hole}} = \left(\frac{k_e Q}{R^2} \right) (\pi r^2) \\
 &= \left(\frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(10.0 \times 10^{-6} \text{ C})}{(0.100 \text{ m})^2} \right) \\
 &\quad \times \pi (1.00 \times 10^{-3} \text{ m})^2 \\
 &= \boxed{28.2 \text{ N} \cdot \text{m}^2 / \text{C}}
 \end{aligned}$$

8. Find the net electric flux through the spherical closed surface shown in Figure P24.8. The two charges on the right are inside the spherical surface.

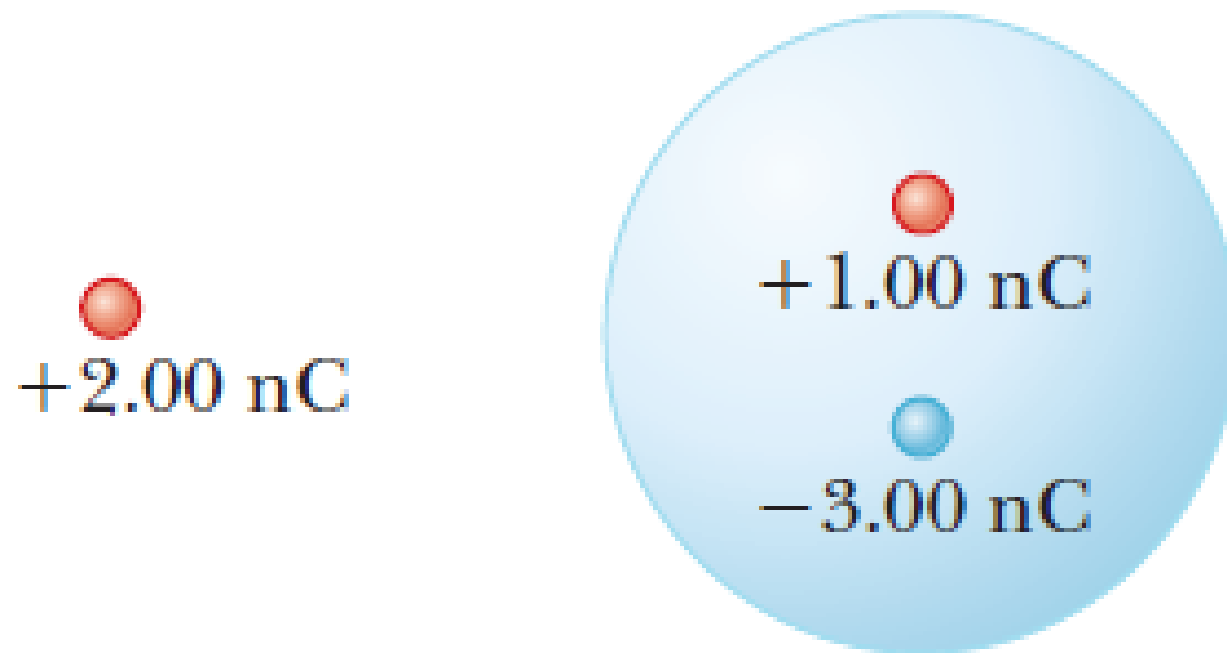
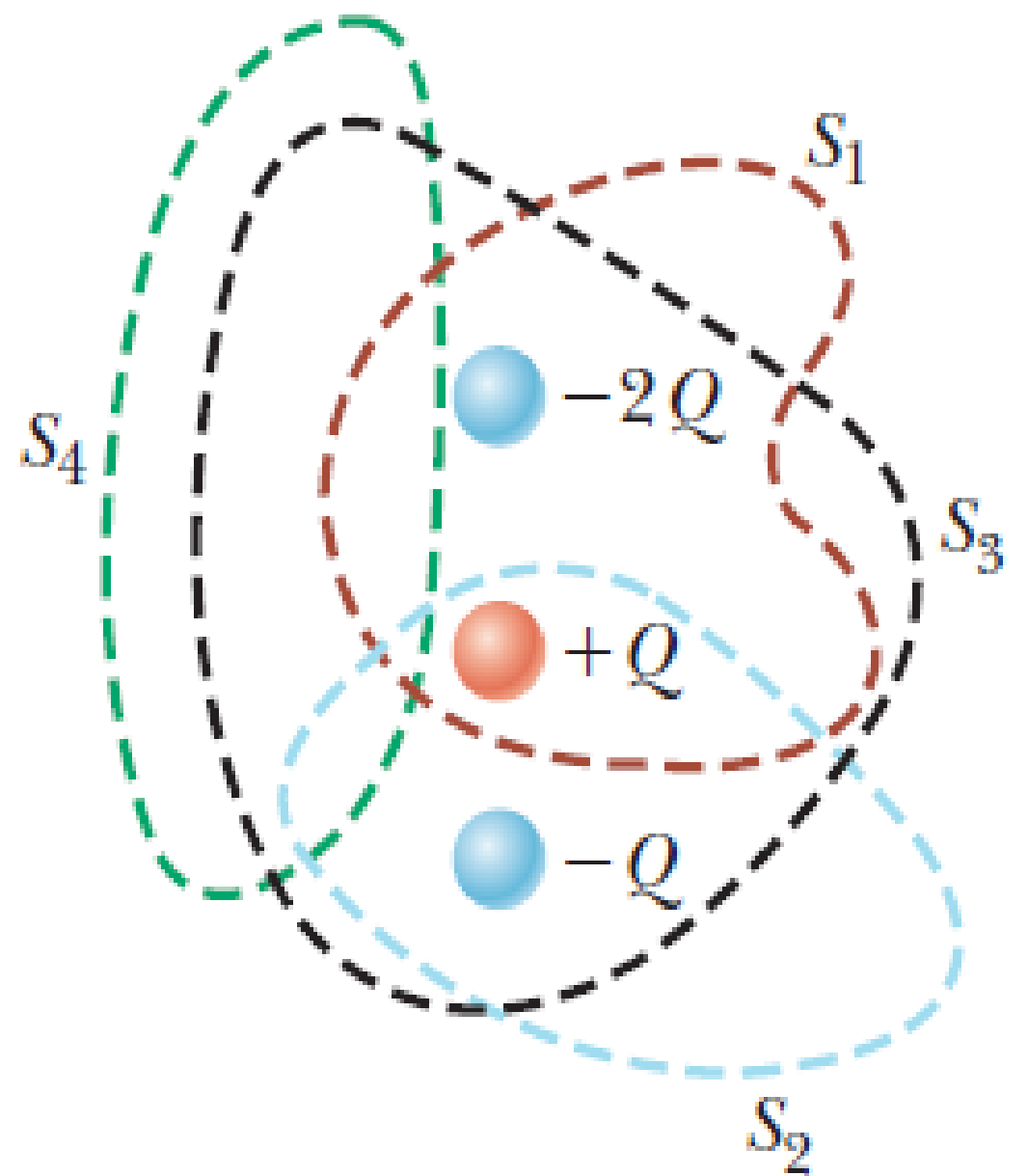


Figure P24.8

$$\Phi_E = \frac{q_{\text{in}}}{\epsilon_0} = \frac{(1.00 \times 10^{-9} \text{ C} - 3.00 \times 10^{-9} \text{ C})}{8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2} = \boxed{-226 \text{ N} \cdot \text{m}^2/\text{C}}$$

11. Four closed surfaces, S_1 through S_4 , together with the charges $-2Q$, Q , and $-Q$ are sketched in Figure P24.11. (The colored lines are the intersections of the surfaces with the page.) Find the electric flux through each surface.



The electric flux through each of the surfaces is given by $\Phi_E = \frac{q_{\text{in}}}{\epsilon_0}$.

Flux through S_1 :

$$\Phi_E = \frac{-2Q + Q}{\epsilon_0} = \boxed{-\frac{Q}{\epsilon_0}}$$

Flux through S_2 :

$$\Phi_E = \frac{+Q - Q}{\epsilon_0} = \boxed{0}$$

Flux through S_3 :

$$\Phi_E = \frac{-2Q + Q - Q}{\epsilon_0} = \boxed{-\frac{2Q}{\epsilon_0}}$$

Flux through S_4 :

$$\Phi_E = \boxed{0}$$

12. A charge of $170\ \mu\text{C}$ is at the center of a cube of edge $80.0\ \text{cm}$. No other charges are nearby. (a) Find the flux through each face of the cube. (b) Find the flux through the whole surface of the cube. (c) **What If?** Would your answers to either part (a) or part (b) change if the charge were not at the center? Explain.

$$\Phi_E = \frac{q_{\text{in}}}{\epsilon_0} = \frac{170 \times 10^{-6} \text{ C}}{8.85 \times 10^{-12} \text{ C}^2 / \text{N} \cdot \text{m}^2} = 1.92 \times 10^7 \text{ N} \cdot \text{m}^2 / \text{C}$$

(a) By symmetry, the flux through each face of the cube is the same.

$$(\Phi_E)_{\text{one face}} = \frac{1}{6} \Phi_E = \frac{1}{6} \frac{q_{\text{in}}}{\epsilon_0}$$

$$\begin{aligned} (\Phi_E)_{\text{one face}} &= \frac{1}{6} \left(\frac{170 \times 10^{-6} \text{ C}}{8.85 \times 10^{-12} \text{ C}^2 / \text{N} \cdot \text{m}^2} \right) \\ &= \boxed{3.20 \times 10^6 \text{ N} \cdot \text{m}^2 / \text{C}} \end{aligned}$$

$$(b) \quad \Phi_E = \frac{q_{\text{in}}}{\epsilon_0} = \left(\frac{170 \times 10^{-6} \text{ C}}{8.85 \times 10^{-12} \text{ C}^2 / \text{N} \cdot \text{m}^2} \right) = \boxed{1.92 \times 10^7 \text{ N} \cdot \text{m}^2 / \text{C}}$$

19. A particle with charge $Q = 5.00 \mu\text{C}$ is located at the center of a cube of edge $L = 0.100 \text{ m}$. In addition, six other identical charged particles having $q = -1.00 \mu\text{C}$ are positioned symmetrically around Q as shown in Figure P24.19. Determine the electric flux through one face of the cube.

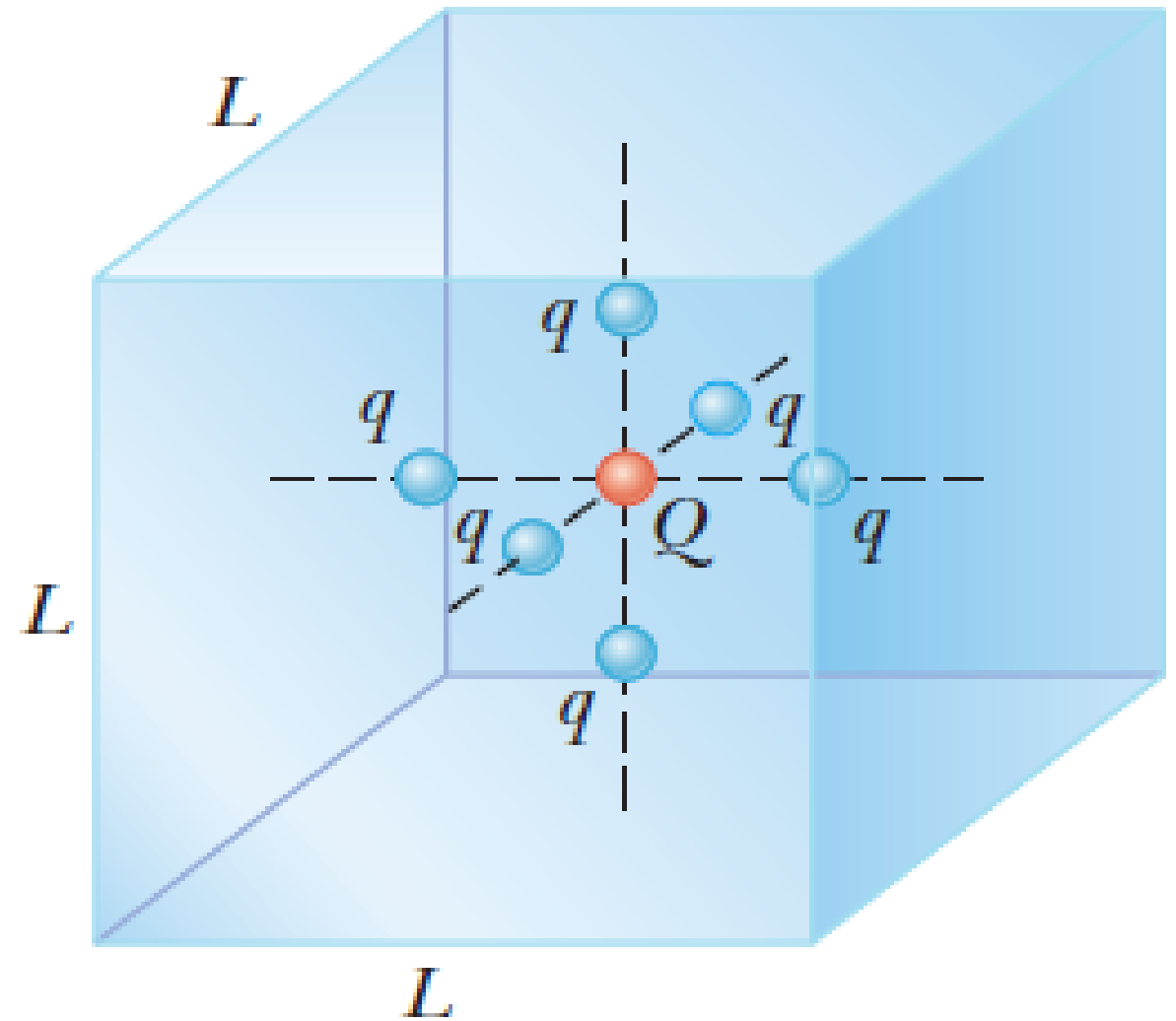


Figure P24.19
Problems 19 and 20.

$$(\Phi_E)_{\text{one face}} = \frac{Q - 6|q|}{6 \epsilon_0}$$

$$(\Phi_E)_{\text{one face}} = \frac{Q - 6|q|}{6 \epsilon_0} = \frac{(5.00 - 6.00) \times 10^{-6} \text{ C} \cdot \text{N} \cdot \text{m}^2}{6 \times 8.85 \times 10^{-12} \text{ C}^2}$$

$$= \boxed{-18.8 \text{ kN} \cdot \text{m}^2 / \text{C}}$$

24. The charge per unit length on a long, straight filament **W** is $-90.0 \mu\text{C}/\text{m}$. Find the electric field (a) 10.0 cm, (b) 20.0 cm, and (c) 100 cm from the filament, where distances are measured perpendicular to the length of the filament.

$$E = \frac{2k_e\lambda}{r}$$

(a) At $r = 10.0 \text{ cm} = 0.100 \text{ m}$,

$$\begin{aligned} E &= \frac{2k_e\lambda}{r} = \frac{2(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(90.0 \times 10^{-6} \text{ C}/\text{m})}{0.100 \text{ m}} \\ &= \boxed{16.2 \text{ MN/C}} \end{aligned}$$

(b) At $r = 20.0 \text{ cm} = 0.200 \text{ m}$,

$$E = \frac{2k_e\lambda}{r} = \frac{2(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(90.0 \times 10^{-6} \text{ C/m})}{0.200 \text{ m}}$$
$$= \boxed{8.09 \text{ MN/C}}$$

(c) At $r = 100 \text{ cm} = 1.00 \text{ m}$,

$$E = \frac{2k_e\lambda}{r} = \frac{2(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(90.0 \times 10^{-6} \text{ C/m})}{1.00 \text{ m}}$$
$$= \boxed{1.62 \text{ MN/C}}$$

25. A 10.0-g piece of Styrofoam carries a net charge of **AMT** $-0.700 \mu\text{C}$ and is suspended in equilibrium above the center of a large, horizontal sheet of plastic that has a uniform charge density on its surface. What is the charge per unit area on the plastic sheet?

$$mg = qE = q \left(\frac{\sigma}{2 \epsilon_0} \right) = q \left(\frac{Q/A}{2 \epsilon_0} \right)$$

$$\frac{Q}{A} = \frac{2 \epsilon_0 mg}{q}$$

$$= \frac{2(8.85 \times 10^{-12} \text{ C}^2 / \text{N} \cdot \text{m}^2)(10.0 \times 10^{-3} \text{ kg})(9.80 \text{ m/s}^2)}{|-0.700 \times 10^{-6} \text{ C}|}$$

$$= \boxed{2.48 \mu\text{C/m}^2}$$

27. A large, flat, horizontal sheet of charge has a charge **M** per unit area of $9.00 \mu\text{C}/\text{m}^2$. Find the electric field just above the middle of the sheet.

$$E = \frac{\sigma}{2\epsilon_0} = 2\pi k_e \sigma$$
$$= (2\pi)(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(9.00 \times 10^{-6} \text{ C}/\text{m}^2)$$

$$\vec{E} = 5.08 \times 10^5 \text{ N/C } \hat{j}$$

29. Consider a thin, spherical shell of radius 14.0 cm with a total charge of $32.0 \mu\text{C}$ distributed uniformly on its surface. Find the electric field (a) 10.0 cm and (b) 20.0 cm from the center of the charge distribution.

(a) A gaussian sphere concentric with the shell, with radius 10.0 cm, encloses 0 charge. $\vec{E} = \boxed{0}$.

(b) $\oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0}$. The field is radially outward, and $4\pi r^2 E = q/\epsilon_0$:

$$E = \frac{k_e q}{r^2} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(32.0 \times 10^{-6} \text{ C})}{(0.200 \text{ m})^2}$$
$$= 7.19 \times 10^6 \text{ N/C}$$

$$\vec{E} = \boxed{7.19 \text{ MN/C radially outward}}$$

30. A nonconducting wall carries charge with a uniform **W** density of $8.60 \mu\text{C}/\text{cm}^2$. (a) What is the electric field 7.00 cm in front of the wall if 7.00 cm is small compared with the dimensions of the wall? (b) Does your result change as the distance from the wall varies? Explain.

(a) The charge per unit area of the wall is

$$\sigma = (8.60 \times 10^{-6} \text{ C/cm}^2) \left(\frac{100 \text{ cm}}{\text{m}} \right)^2 = 8.60 \times 10^{-2} \text{ C/m}^2$$

The electric field at a distance of 2.00 cm is then

$$\begin{aligned} E &= \frac{\sigma}{2 \epsilon_0} = \frac{8.60 \times 10^{-2} \text{ C/m}^2}{2(8.85 \times 10^{-12} \text{ C}^2 / \text{N} \cdot \text{m}^2)} \\ &= \boxed{4.86 \times 10^9 \text{ N/C away from the wall}} \end{aligned}$$

(b) So long as the distance from the wall is small compared to the width and height of the wall, the distance does not affect the field.

35. A solid sphere of radius 40.0 cm has a total positive **W** charge of $26.0 \mu\text{C}$ uniformly distributed throughout its volume. Calculate the magnitude of the electric field (a) 0 cm, (b) 10.0 cm, (c) 40.0 cm, and (d) 60.0 cm from the center of the sphere.

(a) At the center of the sphere, the total charge is zero, so

$$E = \frac{k_e Q r}{a^3} = \boxed{0}$$

(b) At a distance of 10.0 cm = 0.100 m from the center,

$$\begin{aligned} E &= \frac{k_e Q r}{a^3} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C})(26.0 \times 10^{-6} \text{ C})(0.100 \text{ m})}{(0.400 \text{ m})^3} \\ &= \boxed{365 \text{ kN/C}} \end{aligned}$$

- (c) At a distance of 40.0 cm = 0.400 m from the center, all of the charge is enclosed, so

$$E = \frac{k_e Q}{r^2} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C})(26.0 \times 10^{-6} \text{ C})}{(0.400 \text{ m})^2}$$
$$= \boxed{1.46 \text{ MN/C}}$$

- (d) At a distance of 60.0 cm = 0.600 m from the center,

$$E = \frac{k_e Q}{r^2} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C})(26.0 \times 10^{-6} \text{ C})}{(0.600 \text{ m})^2}$$
$$= \boxed{649 \text{ kN/C}}$$

The direction for each electric field is $\boxed{\text{radially outward}}$.

37. A long, straight metal rod has a radius of 5.00 cm and a charge per unit length of 30.0 nC/m. Find the electric field (a) 3.00 cm, (b) 10.0 cm, and (c) 100 cm from the axis of the rod, where distances are measured perpendicular to the rod's axis.

M



$\oint E dA = E(2\pi rl) = \frac{q_{\text{in}}}{\epsilon_0} \quad E = \frac{q_{\text{in}}/l}{2\pi \epsilon_0 r} = \frac{\lambda}{2\pi \epsilon_0 r}$ for the field outside the metal rod.

(a) At $r = 3.00$ cm, $\vec{E} = \boxed{0}$

(b) At $r = 10.0$ cm,

$$\begin{aligned}\vec{E} &= \frac{30.0 \times 10^{-9} \text{ C}}{2\pi(8.85 \times 10^{-12} \text{ C}^2 / \text{N} \cdot \text{m}^2)(0.100 \text{ m})} \\ &= \boxed{5\,400 \text{ N/C, outward}}\end{aligned}$$

(c) At $r = 100$ cm,

$$\begin{aligned}\vec{E} &= \frac{30.0 \times 10^{-9} \text{ C}}{2\pi(8.85 \times 10^{-12} \text{ C}^2 / \text{N} \cdot \text{m}^2)(1.00 \text{ m})} \\ &= \boxed{540 \text{ N/C, outward}}\end{aligned}$$

43. Two identical conducting spheres each having a radius of 0.500 cm are connected by a light, 2.00-m-long conducting wire. A charge of $60.0 \mu\text{C}$ is placed on one of the conductors. Assume the surface distribution of charge on each sphere is uniform. Determine the tension in the wire.

AMT

$$\begin{aligned} F &= \frac{k_e (Q/2)(Q/2)}{(L + R + R)^2} = \frac{k_e Q^2}{4(L + 2R)^2} \\ &= \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(60.0 \times 10^{-6} \text{ C})^2}{4(2.01 \text{ m})^2} = \boxed{2.00 \text{ N}} \end{aligned}$$

44. A square plate of copper with 50.0-cm sides has no net charge and is placed in a region of uniform electric field of 80.0 kN/C directed perpendicularly to the plate. Find (a) the charge density of each face of the plate and (b) the total charge on each face.

(a) $E = \frac{\sigma}{\epsilon_0}$, so

$$\begin{aligned}\sigma &= (8.00 \times 10^4 \text{ N/C})(8.85 \times 10^{-12} \text{ C}^2 / \text{N} \cdot \text{m}^2) \\ &= 7.08 \times 10^{-7} \text{ C/m}^2\end{aligned}$$

$$\sigma = \boxed{708 \text{ nC/m}^2}, \text{ positive on one face and negative on the other.}$$

(b) $\sigma = \frac{Q}{A}$, so

$$\begin{aligned}Q &= \sigma A = (7.08 \times 10^{-7} \text{ C/m}^2)(0.500 \text{ m})^2 \\ &= 1.77 \times 10^{-7} \text{ C} = \boxed{177 \text{ nC}}\end{aligned}$$

positive on one face and negative on the other.